REFERENCE AUDIT

Consolidated reference suggestions

Carmichael intersection Lucas-Carmichael: a four-prime exclusion and a general 10^24 bound

Prepared 8 November 2026

Executive outcome

The bibliography is fundamentally sound: all 28 works are genuine, every bibliography entry is used, and no printed DOI was found to be incorrect. The current version should still be revised before submission because six citation or claim issues could attract referee criticism, while several computational and web references need stronger provenance.

28	28/28	0	6
works verified	entries cited	bad printed DOIs	must-fix areas

Bottom line Do not submit the present reference section unchanged. After the attribution, scope, versioning, and reproducibility corrections summarized here, it should be substantially submission-ready.

1. Priority action map

What must change, what should be strengthened, and what is merely stylistic.

Priority	References	Action
MUST FIX	[11], [13], [16], [19], [23]-[24], [28]	Correct dating or attribution, narrow overbroad claims, replace a dead link, and separate the 10^22 and 10^24 results.
STRENGTHEN	[12], [20]-[22], [25]-[26]	Use archival links, pin versions, and add provenance plus an integrity hash for the computational dataset.
MINOR CLEANUP	[1]-[10], [14]-[15], [17]-[18], [27]	Add issue numbers, final preprint versions, standardized journal titles, page counts, and consistent punctuation.

Strongest overall recommendation

Prefer peer-reviewed or institutional sources already present in the bibliography. In particular, cite [5] instead of relying on the Gilchrist webpage for the 2^64 Baillie-PSW statement, and cite [24] alone for the 10^24 Carmichael tabulation.

Checks already passed

PASS All 28 citation keys resolve to bibliography entries.

PASS Every bibliography entry is cited in the manuscript; there are no unused entries.

PASS The journal papers and preprints are real; no fabricated source was found.

PASS Every DOI currently printed in the manuscript resolves to the intended work.

PASS The principal historical and mathematical attributions are supported by the cited sources.

2. Must-fix citation and claim issues

These changes reduce the greatest risk of a referee challenging the literature review.

[11] Gilchrist - incorrect dating

Problem. The phrase 'reporting the 2009 verification ... below 2^64' is wrong. The 2009 bound was 10^17; the 2^64 computation belongs to the 2013 Feitsma table and subsequent testing.

Recommended action. Preferably remove [11] and cite the already-listed peer-reviewed [5], which explicitly reports that none of the relevant base-2 pseudoprimes below 2^64 passes the Lucas test. If [11] is retained, date it 2013 and remove the 2009 attribution.

Primary source. Baillie-Fiori-Wagstaff, DOI 10.1090/mcom/3616

[13] Grantham - the $620 problem is misattributed

Problem. Grantham's journal article contains a separate $6.20 Frobenius challenge. The $620 problem is the Pomerance-Selfridge-Wagstaff challenge, discussed by Chen and Greene and hosted on Grantham's website.

Recommended action. Replace 'Grantham's $620 problem' with 'the Pomerance-Selfridge-Wagstaff $620 problem, as described by Chen and Greene.' Use [6] for the statement. If the Grantham article remains, separate it from the problem webpage and add DOI 10.1090/S0025-5718-00-01197-2.

Primary source. Grantham journal article

[16] Lenstra-Pomerance - dead and misdescribed URL

Problem. The printed HTTP PDF link is no longer dependable and should not be described as an 'archived copy.'

Recommended action. Cite the official AIM HTML note and optionally the workshop record. The note directly supports the p = 3 (mod 80) statement. Omit page numbers when using the unpaginated HTML version.

Primary source. Official AIM note | Workshop record

2. Must-fix issues - continued

[19] Phatak - claim is too broad

Problem. The manuscript says that no subset of the 2,030-prime set is a Williams number. The report's pruning argument assumes an odd number of selected primes, and the report is unrefereed.

Recommended action. Use: 'an unrefereed report claims that no odd-cardinality candidate subset of the Grantham-Alford 2,030-prime list is a Williams number.' Attribute the list explicitly to Grantham-Alford rather than leaving it to appear as a Chen-Greene construction.

Primary source. viXra:2007.0102v1

[23]-[24] Shallue-Webster - separate the 10^22 and 10^24 bounds

Problem. [23], Advances in tabulating Carmichael numbers, establishes the tabulation through 10^22. The 10^24 result and count 308,279,939 are in [24], specifically the later preprint version.

Recommended action. State that [23] reaches 10^22 and [24] reaches 10^24. Cite [24] as arXiv:2506.09903v3 [math.NT] (submitted 2025; revised 13 January 2026), DOI 10.48550/arXiv.2506.09903.

Primary source. Published 10^22 paper | 10^24 preprint, current version

[28] Zhang - prize needs a separate citation and narrower wording

Problem. The journal article supports the counting heuristic and a specified set of 1,029 primes, but it does not document the later prize. The prize is not an unrestricted reward for the first Williams number.

Recommended action. Say that Zhang developed counting heuristics in [28] and later offered $500 for a Williams number formed from his specified 1,029-prime set. Add the author's challenge page as a separate reference.

Primary source. Published paper | $500 challenge page

Recommended replacement sentence for the tabulation paragraph	Carmichael numbers have been tabulated completely to 10^21 by Pinch [21], independently to 10^22 by Goutier [12] and Shallue-Webster [23], and to 10^24 by Shallue-Webster [24].

3. Reproducibility and link resilience

The underlying works are real; these changes make the citations durable and independently checkable.

[25] carm10e24.bin - add provenance and checksum

The binary is a PARI/GP conversion of the Shallue-Webster data, not an independent proof of completeness. Keep completeness attributed to [24]. Add the conversion date, exact size, and SHA-256:

Date	7 March 2026
Size	12,296,962,769 bytes
SHA-256	`964abcc780b9786ae9f18d75a688cfc83f2bf74fb592ead9f58fd94bf6441946`
Provenance	conversion announcement | file and hash announcement

Reference	Recommended improvement
[26] Supplement	Replace the relative repository path with an immutable version: version 2, 2 November 2026, repository tag v1.1. Use the tagged file or a version-specific archive DOI.
[20] Pinch search	Replace the personal HTTP preprint with the official TUCS volume. Cite it for the 10^16 odd-factor search, not for the 10^21 Carmichael tabulation.
[21] Pinch tables	Use the author-deposited Zenodo dataset. This is the correct source for completeness through 10^21.
[22] Pomerance	Replace the mirror with the CWI archival record or Pomerance's author copy. Expand the container title if space permits.
[12] Goutier	Distinguish the dates: the numerical result was reported 28 December 2022; the readme and data archive are dated 2024. Add direct OEIS-hosted readme and data links.

4. Reference-by-reference ledger

A compact action record for all 28 bibliography entries. 'Verified' means no substantive correction is needed.

Ref.	Status	Work	Recommendation
[1]	Verified	Al-Rasasi-Ghanmi	Metadata and claim support are correct. Add issue no. 2 for completeness.
[2]	Verified	Agrawal-Kayal-Saxena	Correct. Prefer Ann. of Math. (2) 160 (2004), no. 2, 781-793. The 2019 erratum does not affect the cited passage.
[3]	Verified	Alford-Granville-Pomerance	Correct. Add series and issue: Ann. of Math. (2) 139 (1994), no. 3, 703-722.
[4]	Verified	Bouallegue-Echi-Pinch	Correct and supports the general a-Williams attribution. Add issue no. 2.
[5]	Verified	Baillie-Fiori-Wagstaff	Correct. Add issue no. 330. Use this peer-reviewed paper for the no-counterexample-below-2^64 statement.
[6]	Verified	Chen-Greene	Correct. Add issue no. 4 and the official Fibonacci Quarterly article URL. The paper supports the weaker order conditions and the related 2,030-prime divisibilities.
[7]	Verified	Chernick	Correct. Add issue no. 4.
[8]	Minor	Chick	Use arXiv:0711.2915v2 [math.NT] (2008), or 'submitted 2007; revised 2008,' and DOI 10.48550/arXiv.0711.2915.
[9]	Minor	Echi	Standardize venue as Math. Rep. Acad. Sci. R. Soc. Can. 29 (2007), no. 2, 41-47. The terminology claim is supported.
[10]	Minor	Erdos	Add no. 3-4 and DOI 10.5486/PMD.1956.4.3-4.16. The subset-product attribution is fair.
[11]	Must fix	Gilchrist	Remove in favor of [5], or correct the date to 2013 and cite the Feitsma-Galway/CECM data. Delete the false 2009-to-2^64 wording.
[12]	Strengthen	Goutier	Clarify: result reported in 2022; readme/data archived in 2024. Add direct readme and archive URLs.
[13]	Must fix	Grantham	Split the journal paper from the $620 problem webpage. Do not call it 'Grantham's $620 problem.' Add DOI 10.1090/S0025-5718-00-01197-2.
[14]	Verified	Howe	Correct and supports the order-2 Carmichael criterion. Add issue no. 232; arXiv:math/9812089 is optional.
[15]	Verified	Loh-Niebuhr	Correct. Add issue no. 214.
[16]	Must fix	Lenstra-Pomerance	Replace the broken PDF URL with the official AIM HTML note and workshop record; do not call the old URL an archived copy.
[17]	Verified	McIntosh	Correct. Add 'Paper A59, 9 pp.' and the official journal PDF.
[18]	Verified	McIntosh-Dipra	Fully correct. The technical claims and the 10^24 four-prime exclusion are supported. Full author name: Mitra Dipra.
[19]	Must fix	Phatak	Retain the unrefereed warning, restrict the claim to odd-cardinality candidate subsets, and attribute the prime list explicitly to Grantham-Alford.
[20]	Strengthen	Pinch search	Metadata correct. Replace the personal URL with the official TUCS volume. Use only for the 10^16 odd-factor search, not for tabulation to 10^21.
[21]	Strengthen	Pinch tables	Use the TUCS paper plus Zenodo dataset DOI 10.5281/zenodo.13755236. This is the proper 10^21 tabulation source.
[22]	Strengthen	Pomerance	Work and claim support are correct. Replace the mirror with CWI or the author PDF and expand the container title.

Ref.	Status	Work	Recommendation
[23]	Must fix	Shallue-Webster 2025	Metadata correct, but its scope is 10^22. Do not cite it as proof of the 10^24 tabulation.
[24]	Must fix	Shallue-Webster preprint	Cite arXiv:2506.09903v3 (revised 13 January 2026), DOI 10.48550/arXiv.2506.09903. This is the 10^24 source.
[25]	Strengthen	Stamm-Wilbrandt binary	Add conversion provenance, 12,296,962,769-byte size, SHA-256, and mailing-list archive links. The file does not independently prove completeness.
[26]	Strengthen	Vakil supplement	Replace the mutable repository-relative path with version 2, 2 November 2026, tag v1.1 or a version-specific archive record.
[27]	Verified	Williams	Correct. Add issue no. 1 if desired. It supports the existence question, three-prime exclusion, factor-3 exclusion, and original parity conclusion.
[28]	Must fix	Zhang	Published citation is correct; give full name Zhenxiang Zhang and issue 84(291). Cite the later $500 challenge separately and state its 1,029-prime restriction.

Recommended replacement citation records

Ref.	Suggested record
[13]	J. Grantham, *Frobenius pseudoprimes*, Math. Comp. 70 (2001), no. 234, 873-891, doi:10.1090/S0025-5718-00-01197-2. Cite the $620 challenge webpage separately if retained.
[16]	H. W. Lenstra Jr. and C. Pomerance, *Remarks on Agrawal's Conjecture*, AIM workshop note, March 2003; official HTML note and workshop record.
[21]	R. G. E. Pinch, *The Carmichael numbers up to 10^21*, TUCS General Publication 46 (2007), 129-131; associated dataset, Zenodo, doi:10.5281/zenodo.13755236.
[24]	A. Shallue and J. Webster, *Algorithms for Carmichael numbers*, arXiv:2506.09903v3 [math.NT] (submitted 2025; revised 13 January 2026), doi:10.48550/arXiv.2506.09903.
[25]	H. Stamm-Wilbrandt, `carm10e24.bin`, PARI/GP binary conversion of the Shallue-Webster data, 7 March 2026, 12,296,962,769 bytes; include the SHA-256 printed in Section 3.
[26]	R. Vakil, *Exploratory note: two-sided subset products for Williams numbers*, version 2, 2 November 2026, unpublished supplement, repository tag v1.1.
[28a]	Z. Zhang, *A Conjecture on Williams Numbers*, author challenge page; $500 challenge restricted to the specified 1,029-prime set.

5. Ready-to-use manuscript wording

These formulations address the highest-risk attribution and scope issues without changing the mathematical argument.

Baillie-PSW range	*Statement (b) adds little to the known absence of Baillie-PSW counterexamples below 2^64 [5], but it bounds a construction rather than a tested range.*
$620 problem	*The construction attempted by Chen and Greene [6] for the Pomerance-Selfridge-Wagstaff $620 problem uses weaker order conditions.*
2,030-prime report	*The related Grantham-Alford set of 2,030 primes satisfies the two divisibility systems, and an unrefereed report [19] claims that no odd-cardinality candidate subset is a Williams number.*
Tabulation history	*Carmichael numbers have been tabulated completely to 10^21 by Pinch [21], independently to 10^22 by Goutier [12] and Shallue-Webster [23], and to 10^24 by Shallue-Webster [24].*
Zhang challenge	*Zhang [28] developed counting heuristics for Williams numbers formed from a fixed prime set; he later offered $500 for a Williams number formed from his specified set of 1,029 primes [challenge page].*

Submission checklist

[ ]	Correct [11], [13], [16], [19], [23]-[24], and [28].
[ ]	Separate published articles from challenge webpages.
[ ]	Attribute the 10^22 and 10^24 tabulations to the correct sources.
[ ]	Pin the exact version of every preprint and unpublished supplement.
[ ]	Add provenance, byte count, and SHA-256 for the computational dataset.
[ ]	Replace fragile personal-server URLs with institutional, archival, DOI, or tagged-repository links.
[ ]	Add issue numbers and final-version metadata consistently.
[ ]	Rebuild the manuscript PDF and test every hyperlink.
[ ]	Check consistent journal titles, punctuation, dates, and page/article numbering.
[ ]	Adjust the final typesetting if desired: the bibliography is readable, but the author address currently sits on an almost empty final page.

Overall assessment	No source appears fabricated, the printed DOIs are sound, and the principal mathematical citations are well chosen. Once the attribution, scope, versioning, and reproducibility items above are corrected, the reference section should be substantially submission-ready.

Scope note: this report consolidates the bibliographic and claim-support audit of the 28-entry reference section. It does not alter the manuscript and is not a substitute for the target journal's house style.